Cubic gluing beyond 3-torsion: a module recognition theorem

2026-10-25

The new result in this note is a derived linear-algebra classification, with publication priority unresolved. It supplies a checkable kernel criterion for primes beyond three. It does **not** establish a novel affected curve family, construct a quotient curve, or evaluate a discrete-logarithm transfer. The user's acceptance criterion is novelty plus a proof and an explicit transfer; beating rho is not a gate.

Requirement	Status at preregistration
Recognition theorem beyond the previous 3-test	Written proof below; Rust implemented, new runs pending
Exact small-field source set	Complete d=3 census specified; pending
Published-priority audit	Partial; Tian's 2023 full article remains unread
New descended curve and subgroup map	Not constructed
Lean proof of the full classification	Not formalized; fixed constructive case attempted separately

The module theorem

Let ℓ be an odd prime with $\ell \equiv 2 \pmod 3$, and let $k = \mathbb{F}_\ell$. Let $q \in k^*$ and $M \in \mathrm{GL}_2(k)$ satisfy $\det M = q^3$. On $V = (k^2)^3$ set

$$F(v_0, v_1, v_2) = (Mv_2, v_0, v_1), \qquad \Omega = J \oplus qJ \oplus q^2 J,$$

where $J((x,y),(x',y')) = xy' - yx'$. Then F is a similitude with multiplier q. There exists an F-stable Lagrangian L such that $L \cap V_i = 0$ for all three coordinate planes **if and only if** M **is split semisimple over** k: it is either scalar or has two distinct k-rational eigenvalues.

This concerns geometric torsion coordinates transported from a symplectic basis on an elliptic curve over $\mathbb{F}_{q^3}$. Here the displayed q means its residue modulo ℓ; the determinant is q^3, not q. The weighted form accounts for the Frobenius transport of the basis.

Proof: two distinct eigenvalues

Write $Me = \alpha e$ and $Mf = \beta f$. Since cubing is a bijection on k, there is a unique $\lambda \in k^*$ with $\lambda^3 = \alpha$. Put

$$v = (e, \lambda^{-1}e, \lambda^{-2}e), \qquad c_i = q^i \lambda^{-i}.$$

Let W consist of triples $(y_0 f, y_1 f, y_2 f)$ with $c_0 y_0 + c_1 y_1 + c_2 y_2 = 0$. Then $Fv = \lambda v$ and W is stable: it is the perpendicular space of v inside the stable beta-eigenspace of F^3. The direct sum $L = kv \oplus W$ has dimension three and is isotropic. All c_i and all e-components of v are nonzero. An element of L supported in only one coordinate therefore has zero e-part, then zero f-part. Thus L is transverse. An explicit basis for W uses coefficient vectors $(c_1, -c_0, 0)$ and $(c_2, 0, -c_0)$.

Proof: scalar case

If $M = \mu I$, choose $\lambda^3 = \mu$. The determinant identity and injectivity of cubing give $q = \lambda^2$. In coordinates $w_i = \lambda^i v_i$, F becomes lambda times cyclic permutation and Ω becomes the standard sum of three J's. Write $w_i = (x_i, y_i)$. Take $x = Sy$, where S is the symmetric three-by-three matrix with zero diagonal and every off-diagonal entry one. Its graph is a Lagrangian, is stable under cyclic permutation, and is transverse: a vector supported in a single coordinate forces its y-entry, and then its x-entry, to vanish.

Proof: irreducible case

If the characteristic polynomial of M is irreducible quadratic, every irreducible factor of the characteristic polynomial of F has even degree. Indeed, if z is a root of such a factor of degree h, then z^3 is a root of the irreducible quadratic, so $\mathbb{F}_{\ell^2}$ is a subfield of $\mathbb{F}_{\ell^h}$ and $2 \mid h$. The characteristic polynomial of an invariant subspace divides that of F. An invariant subspace of odd dimension, in particular dimension three, is impossible.

Proof: nonscalar Jordan case

Write $M = \mu I + N$ with $N^2 = 0$, $N \neq 0$. Cubing bijective gives a unique $\lambda^3 = \mu$. The polynomial $T^3 - \mu$ factors over k into one linear factor and one irreducible quadratic, since k has no nontrivial cube root of unity. The cyclic block companion F has minimal and characteristic polynomial $(T^3 - \mu)^2$: applying a putative annihilator of degree below six to a vector with a cyclic M-vector in one coordinate would give an M-annihilator of degree below two. Equivalently, its module is $k[T]/((T^3 - \mu)^2)$.

Its primary pieces are therefore $k[T]/(T - \lambda)^2$ and $k[T]/h^2$, of dimensions two and four. An invariant three-space must have dimensions one and two in these pieces, respectively. In each piece that submodule is unique. Their sum is exactly $\ker(F^3 - \mu I)$, the direct sum of the same M-eigenline in all three coordinates. It contains a line in each coordinate plane and fails transversality. These four cases exhaust M.

A source-trace screen at ell=5

For $E/\mathbb{F}_{q^3}$ with trace t, put $\Delta = t^2 - 4q^3 \pmod 5$. A nonzero square Delta proves the positive split-distinct case; a nonsquare proves the irreducible case. If Delta vanishes, the trace does not distinguish scalar from Jordan Frobenius. The two matrices I and $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ have the same characteristic polynomial but opposite outcomes.

For ordinary binary curves over F512 relative to F8, the old 3-test is $t = 0 \pmod 3$. Here $512 = 2 \pmod 5$, so the sufficient 5-test becomes $t = 2$ or $3 \pmod 5$. Consequently traces satisfying the latter and $t \neq 0 \pmod 3$ pass a different torsion-kernel test. This does not mean that the source curves were previously unknown to cover attacks.

Geometric boundary

A maximal isotropic kernel gives a principally polarized quotient of the Weil restriction. Transversality alone has not been proved sufficient here for its geometric indecomposability at ell=5. The earlier ell=3 proof uses degree partitions of three; it cannot simply be reused for five. The partitions $5 = 1 + 4 = 2 + 3 = 1 + 1 + 3 = 1 + 2 + 2$ introduce possible geometric isogenies of degrees up to six between source conjugates. Excluding those is a sufficient route to an indecomposability proof, still to be checked on any source promoted beyond this module screen. Arithmetic Torelli, source-twist selection, quotient reconstruction and explicit log transport remain open. No curve ID or catalogue attack edge is issued here.

Prior art and provenance

- Tian, 2020 preprint, section 3 already assumes a rational maximal isotropic kernel avoiding the coordinate elliptic factors and constructs the resulting isogeny in an odd-characteristic model. The method is not new. The exact recognition classification above is not asserted in the inspected section. This does not establish priority.
- Tian, 2023 journal version remains accessible only as a subscription preview in this session. Its full contents have not been compared, so novelty stays unresolved.

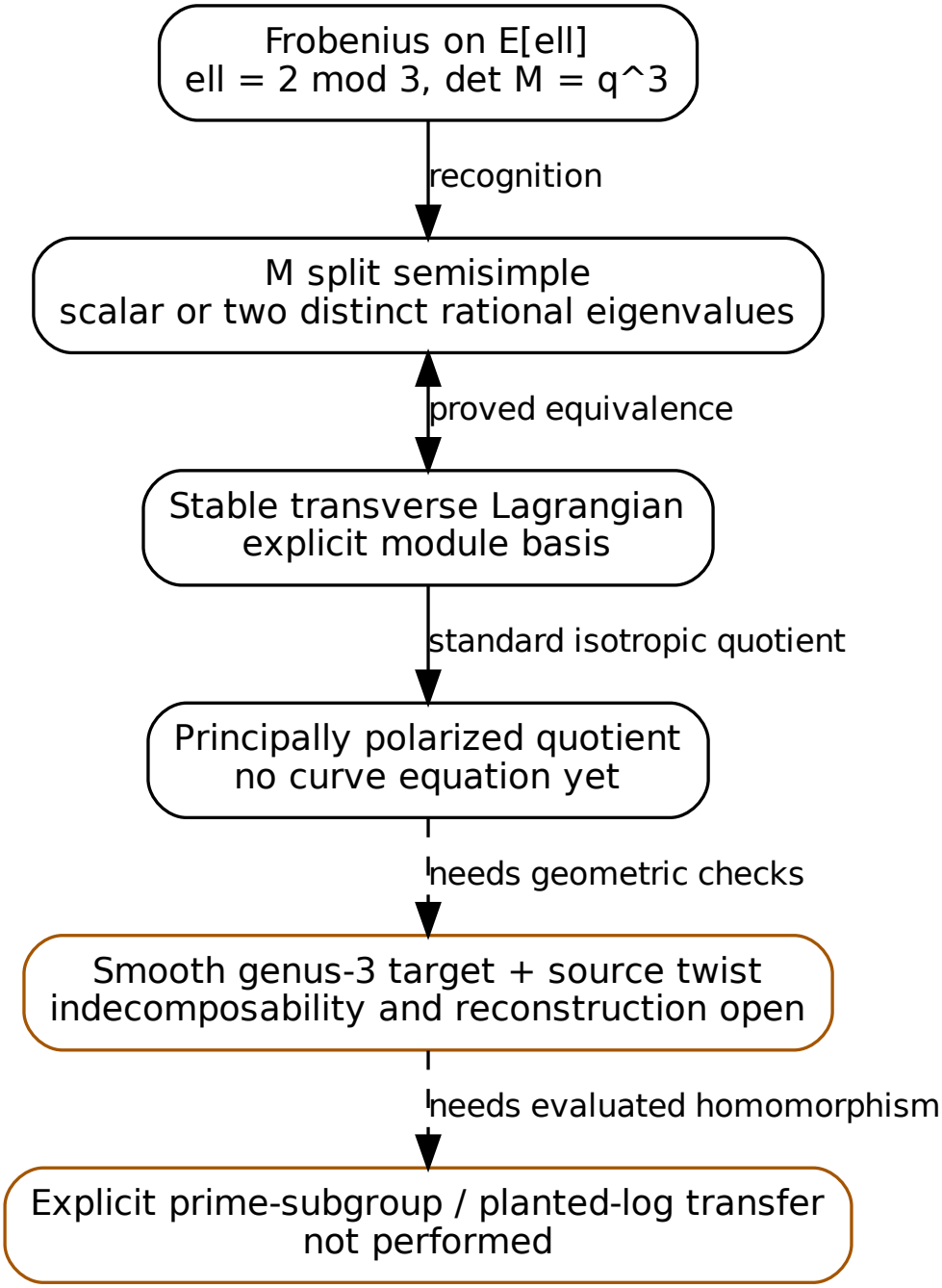


Figure 1: Scope of the recognition result. Solid arrows are proved module implications; dashed arrows are unfinished geometric constructions.

- Tian et al., 2018 explicitly describes an odd-characteristic degree-three cover construction for cofactor-two curves. Its abstract is not a binary reconstruction theorem; full theorem-level comparison is still owed.
- Hess's generalized descent already covers the ordinary full-relative-degree binary sources in the broad existence sense; see the frozen `HESS_SCOPE.md`. A new torsion predicate cannot by itself create a historically new affected family.

Current main was integrated at `dae09d5f` from `c5aa7b39bfa30e8cdbd46a158c7cbadaa8be3503`. Its instructions removed the old missing `experiment.protocol/v1` checker dependency. Earlier reports remain dated snapshots; their mathematical limitations are unchanged. The referenced sibling lead-atlas tool and shared lab-contract file are absent in the available sibling revision, so no atlas review is represented as done.

Exact commands and results are appended in a versioned result note after the frozen checks run. The old kernel and scope diagrams were inspected; they make no ell=5 claim and remain unchanged. Canonical curve/route graphs receive no new edge without a constructed target and checked subgroup map. The remaining target and novelty obligations are tracked in `problems/binary-cubic-gluing-target-and-novelty.md`.